

MADLEY, United KingdomWMO: **035210**Lat: **52.033N**Lon: **2.850W**Elev: **256**StdP: **14.56**Time Zone: **0.00 (EUW)**Period: **86-97**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	24.3	26.9	21.9	16.6	26.5	24.5	18.8	29.6	31.0	44.0	27.6	49.1	1.2	60	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	77.3	65.8	74.3	64.5	71.5	62.8	67.4	74.7	65.4	72.6	63.7	70.0	5.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	92.3	71.1	62.7	86.1	68.9	60.9	80.7	66.9	31.9	74.7	30.4	72.8	29.1	70.1	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 23.3	19.7	17.5	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	17.5	70.0	5.0	1.8	13.9	71.3	11.0	72.4	8.2	73.5	4.6	74.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	41.4	40.2	43.9	46.6	52.1	57.5	61.3	60.0	54.9	50.4	44.6	41.8
	DBStd	9.06	6.02	6.84	5.17	5.11	4.97	4.96	3.77	4.81	4.36	5.63	6.27	7.28
	HDD50	1428	268	277	196	127	34	3	0	0	9	66	183	265
	HDD65	5644	730	695	653	552	401	231	125	170	304	453	612	718
	CDD50	1288	2	2	8	25	99	227	350	309	156	78	20	12
	CDD65	28	0	0	0	0	0	5	10	13	0	0	0	0
	CDH74	289	0	0	0	0	5	70	104	107	3	0	0	0
	CDH80	34	0	0	0	0	0	7	5	22	0	0	0	0
(14) Wind	WSAvg	5.9	7.9	8.3	7.8	6.2	5.2	4.6	4.6	4.4	4.5	5.1	5.5	7.2
(15) Precipitation	PrecAvg	31.8	3.5	2.6	2.0	2.2	2.4	2.1	2.2	2.3	2.4	3.5	3.2	3.4
	PrecMax	41.3	7.2	6.9	3.8	6.2	4.2	5.8	6.5	5.9	5.8	6.9	6.2	6.6
	PrecMin	20.8	0.3	0.2	0.4	0.2	0.2	0.4	0.3	0.4	0.4	0.6	1.5	0.9
	PrecStd	5.0	1.5	1.6	1.0	1.5	0.9	1.4	1.4	1.4	1.4	1.6	1.2	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	55.2	59.5	67.2	74.7	80.4	80.8	81.4	72.9	65.8	60.3	58.1
		MCWB	51.8	51.2	52.9	58.4	62.5	66.5	65.6	66.6	63.2	60.3	56.7	54.1
	2%	DB	53.2	53.5	55.8	61.9	70.7	75.2	77.4	76.4	68.1	62.5	58.2	56.5
		MCWB	49.7	49.8	50.9	54.6	61.0	65.1	65.6	66.3	60.6	58.3	55.1	53.0
	5%	DB	51.7	51.6	53.7	58.5	66.4	71.7	74.3	72.6	65.7	60.3	55.7	54.8
		MCWB	48.9	48.4	49.7	52.6	58.2	62.6	64.5	64.3	59.0	56.8	53.0	51.7
	10%	DB	50.1	49.8	52.0	55.9	62.3	68.0	71.4	69.3	63.2	58.6	54.0	52.5
		MCWB	47.4	46.9	48.2	50.9	55.6	60.6	62.7	62.1	57.5	55.8	51.5	49.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	52.3	54.0	59.4	64.0	68.5	69.5	70.2	64.5	61.4	57.7	54.8
		MCDB	54.4	54.5	57.9	66.5	72.6	76.0	78.2	77.1	70.0	64.3	59.9	57.2
	2%	WB	50.6	50.4	51.8	55.5	61.3	65.7	67.1	67.5	62.3	59.1	55.5	53.5
		MCDB	52.7	52.5	55.0	59.8	69.7	73.9	74.5	74.5	66.7	61.4	57.6	56.0
	5%	WB	49.3	48.8	50.3	53.4	58.9	63.1	65.2	65.4	60.3	57.7	53.7	52.0
		MCDB	51.4	51.2	53.2	58.2	65.2	70.7	72.4	71.8	63.8	59.9	55.4	54.6
	10%	WB	47.8	47.1	49.0	51.5	56.6	61.0	63.6	63.2	58.6	56.3	51.9	50.0
		MCDB	50.0	49.7	51.7	55.4	61.4	67.5	70.1	68.1	62.4	58.5	53.6	52.3
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	9.6	11.4	13.9	15.6	15.4	15.3	15.3	14.7	11.8	10.2	8.2
		MCDBR	10.7	11.1	13.5	19.6	23.8	23.6	23.1	21.0	19.4	13.7	9.9	9.5
		MCWBR	10.1	9.5	10.1	13.0	14.9	13.7	13.0	11.6	12.7	10.5	8.9	8.6
	5% WB	MCDBR	10.5	10.3	12.6	18.3	21.8	21.3	20.0	18.5	15.7	11.8	9.9	9.0
		MCWBR	10.2	9.4	10.2	12.5	14.3	13.5	12.4	11.6	11.7	9.7	9.4	8.3
(40) Clear-Sky Solar Irradiance	taub		0.322	0.338	0.371	0.394	0.400	0.393	0.390	0.375	0.365	0.349	0.323	0.312
	taud		2.339	2.368	2.277	2.227	2.248	2.300	2.311	2.370	2.399	2.415	2.392	2.316
	Ebn at Noon		213	246	258	266	270	273	272	270	259	238	212	194
	Edh at Noon		20	26	34	40	41	39	38	34	30	24	19	17
(44) All-Sky Solar Radiation	RadAvg		238	449	807	1241	1533	1650	1540	1267	958	541	285	184
	RadStd		21	52	100	143	84	178	159	105	73	49	29	22

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page